

Binoy Saha

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EDUCATION

University of Texas at Arlington, Honors College

Data Science (GPA: 3.8)

- **Coursework:** Calculus 1 & 2, Intermediate Programming, Python for Data Science 1 & 2, Data Mining

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, HTML/CSS, Java, SQL
- **Backend:** Node.js, Express, FastAPI, Spring Boot, Restful API, GraphQL, JWT, OAuth
- **Frontend:** React, React Native, Next.js
- **Databases:** PostgreSQL, MySQL, MongoDB
- **DevOps & Cloud:** Docker, Kubernetes, CI/CD, AWS (EC2, S3), Linux, Unix
- **Data:** Apache Kafka, Apache Spark, NumPy, Pandas
- **Tools:** Git, Postman, VSCode, Agile / Scrum, Microservice Architecture

EXPERIENCE

ACM @ UT Arlington | *AI AutoReel - Full-Stack Backend Developer*

Feb 2026

- Deployed a production REST API backend using **FastAPI** and **PostgreSQL**, implementing JWT-based authentication middleware with bcrypt password hashing to serve a **10,000+** user sports highlight generation platform
- Engineered a multi-stage video processing pipeline orchestrating **FFmpeg** audio extraction, **Whisper** ML speech transcription, RMS/dB acoustic spike detection, and linguistic keyword scoring into a single asynchronous job queue, reducing end-to-end highlight generation latency by **40%**
- Designed and shipped a **RESTful API** layer with CORS middleware, token-protected route guards, and structured error handling (HTTP 401/404/500), consumed by a **React.js/Next.js** frontend
- Containerized the full application stack using **Docker** for consistent local and production environments; managed collaborative development across a 6-person team using Git feature branching, pull requests, and code review workflows
- Integrated **Whisper transcription and acoustic spike detection** as a **microservice** in the backend pipeline, enabling automated audio analysis for each highlight and reducing manual review time

State Farm, ACM | *Undergraduate Researcher*

Feb 2025

- Built a **YOLO + RF-DETR** object detection pipeline for UAV-based hail damage surveillance, implementing a complete **COCO-to-YOLO** data preprocessing pipeline handling **10,000+** annotated images
- Benchmarking **CNN vs transformer-attention** architectures (**precision, recall, mAP**) to inform production deployment decisions for autonomous drone systems; co-authoring paper for ACM Multimedia Conference 2026
- Containerized model training environment using **Docker** for reproducible experimentation across team hardware configurations

Office of Information & Technology, UTA | *Frontend Technical Assistant*

Aug 2025

- Authored 25+ **ServiceNow** knowledge base webpages adhering to **WCAG AA** accessibility standards using **HTML/CSS/JavaScript/Bootstrap**, reducing ticket resolution time by **30%** for **10,000+** campus users
- Managed documentation lifecycle using **Git** version control, coordinating with **15+** department stakeholders across distributed IT operations teams

RoboMasters Robotics, UTA | *Autonomous Systems Developer*

Nov 2025

- Selected as only underclassman on competitive robotics team; architected a modular real-time computer vision pipeline in **Python/OpenCV** achieving **30+ FPS** on live video, integrated with **C++ Taproot** embedded framework via inter-process communication
- Implemented target **prioritization algorithm** using contour detection, HSV color space filtering, and confidence scoring, with outputs consumed by servo motor control systems over a defined API contract

TECHNICAL PROJECTS

TechPulse – Neural RAG Media Intelligence Pipeline | [Github](#)

Python, FastAPI, Pinecone, Sentence-Transformers, Claude API

- Built an end-to-end news ingestion and retrieval pipeline using NewsAPI across 6 major outlets, embedding 100+ articles into a **Pinecone** vector store (384-dim, cosine similarity) with a **sentence-transformers** encoding layer for semantic chunking and retrieval
- Engineered a two-stage ranking system replacing standard cosine-only RAG with a neural re-ranking layer powered by **Meta's TRIBE v2** fMRI foundation model, predicting cortical activation across **20,484 brain vertices** to score content by human neural engagement
- Designed and deployed a **FastAPI microservice** exposing the TRIBE inference pipeline as a REST endpoint, enabling modular plug-in re-ranking on top of any retrieval system; integrated **Claude Sonnet** downstream for multi-outlet framing analysis
- Validated re-ranking signal against baseline cosine retrieval across 6 outlets and **50+** retrieved chunks per query, surfacing measurable neural engagement divergence (**+0.0672 vs -0.0338 on identical geopolitical queries**) — demonstrating outlet framing bias undetectable by semantic similarity alone

Stock Price Prediction - Monte Carlo & ML Pipeline | [Github](#)

Python, NumPy, Pandas, scikit-learn, yfinance API

- Built an end-to-end data pipeline: **REST API** data ingestion via **yfinance**, **NumPy/Pandas** feature engineering with 5-day sliding window, **Linear Regression** model achieving **R²=0.95** and **RMSE=\$3.30** on AAPL test data
- Implemented **Monte Carlo simulation with 10,000 stochastic scenarios** to generate confidence intervals, demonstrating probabilistic uncertainty quantification in a production-style **Python** pipeline